

Problem

Compute the greatest common divisor of 84 and 30, then add 4.

(A) Generic + Direct

The gcd of 84 and 30 is 30.
Then add 4, so the answer is 34.
30. 30. 30. 30. 30. ...
34. 34. 34. 34. 34. ...
[repeats and loses structure]

✗ Broken CoT ✗ Corrupted Output

(B) Domain + Direct

Use the Euclidean algorithm.
 $84 = 30 \cdot 2 + 24$
 $30 = 24 + 6$
 $24 = 4 \cdot 6$
So the gcd is 24.
Then $24 + 4 = 28$.
Final answer: 28

⚠ Shallow Reasoning ✗ Incorrect Result

(C) Domain + Stronger Model Generated

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I should compute the gcd first and then add 4. The Euclidean algorithm should make this straightforward, so I will reduce the pair step by step and then add the offset at the end.

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$$84 = 30 \cdot 2 + 24$$

$$30 = 24 + 6$$

$$24 = 6 + 0$$

So the gcd is 6.

Final answer: 6

⚠ Partial Reasoning ⚠ Almost Correct

(D) Domain + Self-Generated

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Okay, so I need to find $\gcd(84, 30)$ first, and then add 4. The safest way is the Euclidean algorithm.

$$84 = 30 \cdot 2 + 24$$

$$30 = 24 + 6$$

$$24 = 6 \cdot 4 + 0$$

So the last nonzero remainder is 6.

Then $6 + 4 = 10$.

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Therefore, $\gcd(84, 30) = 6$, and after adding 4 the result is 10.

Final answer: 10

✓ Stable Reasoning ✓ Correct Result